Chapter 1 - Bernoulli's Equation: Reference

1.1 Bernoulli's equation. For the steady flow of an incompressible, inviscid fluid along a streamline, the following quantity is conserved between any two points 1 and 2 on that streamline:

$$P_1 + (1/2) \cdot \rho \cdot v_1^2 + \rho \cdot g \cdot h_1 = P_2 + (1/2) \cdot \rho \cdot v_2^2 + \rho \cdot g \cdot h_2$$

where P is the static pressure, rho is the fluid density, v is the flow speed, $g = 9.81\ m/s^2$ is gravitational acceleration, and h is the elevation. The term P is the static pressure, $(1/2)\rho v^2$ is the dynamic pressure, and $\rho g h$ is the gravitational (hydrostatic) term.

1.2 Continuity equation. For an incompressible fluid in steady flow, the volumetric flow rate $Q = A \cdot v$ is constant, so for a pipe that changes cross-sectional area:

$$A_1 \cdot v_1 = A_2 \cdot v_2$$

This lets you find an unknown speed from the two areas and the other speed.

1.3 Assumptions. Bernoulli's equation as written above holds only when the flow is (a) steady, (b) incompressible (constant rho), (c) inviscid (no friction losses), and (d) evaluated along a single streamline.

1.4 Special cases.
- Horizontal flow: when $h_1 = h_2$ the gravitational terms cancel and $P_1 + (1/2)\rho v_1^2 = P_2 + (1/2)\rho v_2^2$.
- Torricelli's theorem: for a large open tank with a small hole a depth h below the surface, the surface and the jet are both at atmospheric pressure and the surface speed is negligible, so Bernoulli reduces to $v = \sqrt{2 \cdot g \cdot h}$.
- Venturi meter: a horizontal constriction; combine continuity ($A_1 v_1 = A_2 v_2$) with horizontal Bernoulli to relate the pressure drop to the flow rate $Q = A_1 \cdot v_1$.

1.5 Worked example. Water ($\rho = 1000\ kg/m^3$) flows steadily through a horizontal pipe that narrows from area $A_1 = 0.10\ m^2$ to $A_2 = 0.025\ m^2$. In the wide section $v_1 = 2.0$ m/s and the gauge pressure is $P_1 = 150$ kPa. Find v_2 and P_2.
Step 1 - continuity: $v_2 = A_1 \cdot v_1 / A_2 = (0.10 \cdot 2.0) / 0.025 = 8.0$ m/s.
Step 2 - horizontal Bernoulli ($h_1 = h_2$): $P_2 = P_1 + (1/2)\rho(v_1^2 - v_2^2)$.
Step 3 - substitute: $P_2 = 150000 + 0.5 \cdot 1000 \cdot (2.0^2 - 8.0^2) = 150000 - 30000$
$= 120000$ Pa $= 120$ kPa.
So $v_2 = 8.0$ m/s and $P_2 = 120$ kPa.

Chapter 1 - Bernoulli's Equation: Problems

Problem 1. Water flows through a horizontal pipe that narrows from a cross-sectional area of 0.08 m^2 to 0.02 m^2. The speed of the water in the wider section is 1.5 m/s and the gauge pressure there is 120 kPa. Find the speed and the pressure in the narrower section. Assume the water is incompressible and the flow is steady (density = 1000 kg/m^3).

Problem 2. A large open tank is filled with water to a height of 5.0 m above a small hole in its side. Using Bernoulli's equation, determine the speed at which water exits the hole (Torricelli's theorem). Take g = 9.81 m/s^2.

Problem 3. An ideal fluid of density 800 kg/m^3 flows steadily up a pipe that rises 3.0 m. At the lower point the pressure is 200 kPa and the speed is 2.0 m/s; at the upper point the cross-sectional area is half that of the lower point. Find the gauge pressure at the upper point.

Problem 4. A Venturi meter has a throat area of 0.005 m^2 in a pipe of area 0.020 m^2 carrying water (density 1000 kg/m^3). The measured pressure drop between the wide section and the throat is 15 kPa. Determine the volumetric flow rate of the water through the meter.